AortaClip: Geometric Learning for Automated Aortic Surface Mesh Clipping for Computational Fluid Dynamics

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Abstract. Patient-specific Computational Fluid Dynamics (CFD) can support Transcatheter Aortic Valve Implantation (TAVI) planning, but preparing a simulation-ready aortic model requires clipping the segmented surface at the aortic annulus and brachiocephalic artery (BCA), currently done by hand through slow, operator-dependent landmark placement. We propose *AortaClip*, an automated, landmark-free framework that predicts both clipping planes directly from a segmented aortic mesh. The surface is decomposed into Mesh-SLIC superpixels and represented as a graph with rotation- and translation-invariant node features that depend on shape, not scan position. A Graph Attention Network (GATv2) reasons over this graph, and two rotation-equivariant heads output each plane as an origin (soft-argmax over superpixel centroids) and a normal (weighted wall-normal covariance), avoiding intermediate landmarks. As the input encodes no landmark information, the pipeline applies unchanged to new patients. Trained on 15 patients and evaluated on 5 held-out patients from the Barts BRC cohorts, AortaClip reproduces the expert-clipped geometry to a symmetric mean surface distance of 1.48 mm and Dice overlap of 0.88, with the annulus plane within 5.95 mm; the largest errors occur at the BCA outlet. These results show anatomically faithful clipping can be automated without manual intervention.

Keywords: Transcatheter Aortic Valve Implantation (TAVI); Computational Fluid Dynamics (CFD); Aortic Clipping; Geometric Deep Learning; Graph Attention Network; Aortic Surface Mesh.

1 Introduction

Transcatheter Aortic Valve Implantation (TAVI) is a minimally invasive procedure that replaces a narrowed aortic valve without open-heart surgery, and is now widely used for patients who are too high-risk for conventional surgery. Because the replacement valve must seat correctly within each patient's own anatomy, planning benefits from patient-specific Computational Fluid Dynamics (CFD) simulations, which model the haemodynamics in a patient's aorta and help clinicians anticipate procedural risks before the valve is deployed.

Before such a simulation can run, the aortic surface mesh must be clipped at two anatomically meaningful boundaries: an inlet at the aortic annulus, where the new valve seats, and an outlet at the brachiocephalic artery (BCA), the first branch off the arch. These two cuts define the inlet and outlet of the CFD domain, so placing them accurately is a critical step in preparing the simulation.

In current practice these clipping planes are placed by hand, by marking anatomical landmarks in software such as 3D Slicer. While effective, this is labour-intensive and operator-dependent: different operators place the landmarks slightly differently, and that variability propagates into every CFD geometry built from them. This limits both the reproducibility of a single study and the ability to scale to the large cohorts that clinical research needs. Automating clipping-plane generation would therefore remove this bottleneck.

Recent advances in geometric deep learning have shown Graph Neural Networks (GNNs) to be effective for analysing anatomical surface meshes, particularly for segmentation and landmark localisation. These approaches have, however, largely been built for labelling tasks, assigning a class to each point or region, and do not directly predict the clipping planes needed for CFD preprocessing, leaving the manual clipping step unautomated.

To address this gap, we propose *AortaClip*, a fully automated, landmark-free framework that predicts patient-specific clipping planes directly from a segmented aortic surface mesh. The aortic surface is represented as a superpixel graph whose node features are rotation- and translation-invariant, so the representation depends on shape rather than on scan position. A GATv2 network reasons over this graph, and rotation-equivariant prediction heads output the annulus (inlet) and BCA (outlet) planes directly, each as an origin and a normal, rather than predicting intermediate landmarks. The predicted planes are then applied to the mesh and cleaned into a single CFD-ready vascular segment, with no manual intervention at any stage. The main contributions of this work are:

- A fully automated, landmark-free framework that predicts the annulus and BCA clipping planes directly from a segmented aortic surface mesh, representing the surface as a shape-only superpixel graph so that the same pipeline applies unchanged to a new, unannotated patient.

- A GATv2 graph model with rotation-equivariant heads that output each clipping plane as an origin and a normal read from the patient's own geometry, together with the automated clipping and mesh post-processing that turn those planes into a CFD-ready segment.
- A two-level evaluation against expert references, on the predicted planes and on the resulting clipped meshes, showing close agreement over the bulk of the vessel and identifying the smaller outlet branch as the main source of error.

2 Related Work

Graph Neural Networks. Graph Attention Networks (GAT) [15] introduced an attention mechanism that enables graph neural networks to adaptively weight the importance of neighbouring nodes during feature aggregation, improving the representation of graph-structured data compared with uniform neighbourhood aggregation. GATv2 [3] further extends this formulation by introducing dynamic attention, allowing attention weights to depend jointly on both the query and neighbouring nodes. Owing to their ability to model complex geometric relationships, GAT-based architectures are well suited for learning anatomical structures from vascular graphs and form the backbone of the proposed framework.

Superpixel-Based Graph Representation. SLIC superpixels [1] provide an efficient representation by grouping neighbouring vertices into compact regions while preserving anatomical boundaries. Rather than constructing graphs from individual vertices, superpixel-based representations significantly reduce computational complexity while retaining the underlying vascular geometry. This compact representation forms the basis of the graph construction stage in the proposed pipeline.

Automated CFD Model Preparation. Recent work has investigated AI-driven automation of patient-specific CFD workflows. Du *et al.* [4] proposed an end-to-end pipeline for constructing CFD-ready vascular models from medical images, demonstrating the potential of deep learning to reduce manual preprocessing. Similarly, graph-based methods have successfully represented vascular anatomy for segmentation and classification tasks [16]. Nevertheless, automatic prediction of annulus and brachiocephalic artery clipping planes remains largely unexplored. The proposed framework addresses this gap by predicting patient-specific clipping planes directly from segmented aortic surface meshes.

Computational Modelling for TAVI. Patient-specific computational modelling is increasingly used to study valve biomechanics, haemodynamics, and long-term device durability in transcatheter aortic valve implantation (TAVI). Rauseo *et al.* [14] reviewed the use of finite element analysis (FEA), computational fluid dynamics (CFD), and fluid–structure interaction (FSI) for these purposes, and identified labour-intensive geometry preparation and preprocessing as a major barrier to wider adoption. Du *et al.* [4] likewise motivate an automated model-construction pipeline for aortic CFD by the cost of manual preprocessing,

while Queirós *et al.* [12] report operator dependence in manual aortic annulus measurement from CT, a source of variability that propagates into any downstream geometry defined from those landmarks. These observations motivate the present work: by automating clipping plane prediction from segmented aortic surface meshes, our framework reduces manual intervention in generating CFD-ready vascular geometries for patient-specific TAVI simulations.

3 Method

3.1 Pipeline Overview

As illustrated in Figure 1, the proposed framework performs automatic prediction and reconstruction of the inlet and outlet clipping planes directly from a Computed Tomography Angiography (CTA) scan. The thoracic aorta is first segmented and converted into a triangular surface mesh in the Left–Posterior–Superior (LPS) coordinate system. The mesh undergoes landmark-free preprocessing, including a fixed 260 mm top-down axial crop, followed by Mesh-SLIC superpixel partitioning and graph construction with rotation-invariant node features. The resulting graph is processed by a GATv2 backbone with a global context pooling layer, and two rotation-equivariant heads estimate each plane as an origin and a normal. Finally, the predicted planes are used to cut the aortic surface mesh, producing a CFD-ready vascular geometry for downstream haemodynamic analysis.

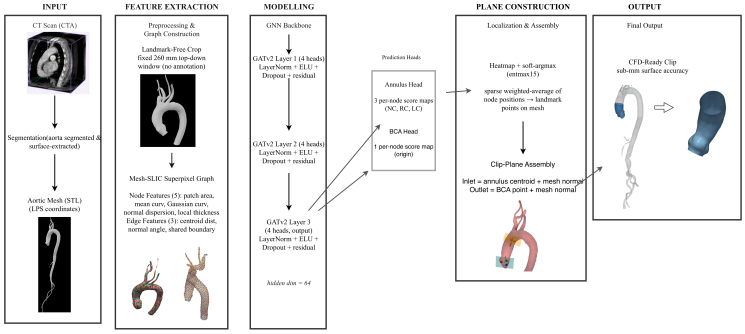


Fig. 1. Overview of the proposed automatic clipping-plane prediction framework.

3.2 Mesh Preprocessing

Each patient's segmented thoracic aorta is exported as a triangular surface mesh in a common physical frame (Left–Posterior–Superior (LPS) anatomical coordinate system, millimetres). To standardise the field of view without any annotation, the mesh is cropped in a purely landmark-free manner: a fixed axial window of height 240 mm is taken from the top of the mesh (with a 5 mm margin), which reliably contains the aortic root and arch across the cohort, where the annulus lies 150–236 mm below the mesh apex. The lateral extent is bounded by the points within this window, expanded by a 5 mm margin. After clipping, only the largest connected component is retained to remove spurious fragments,

and the surface is re-triangulated and cleaned. Because the crop is defined solely by the mesh's own extent, no target information enters the input, and the same procedure applies unchanged to a new, unannotated patient.

3.3 Mesh-SLIC Superpixel Construction

The cropped surface is over-segmented into compact, boundary-adherent superpixels by a mesh adaptation of SLIC [1]. The number of superpixels is set adaptively from the mesh resolution, targeting one superpixel per 50 vertices and clamped to $[200, 3000]$, which keeps superpixel size, and hence the attainable landmark resolution, constant despite varying mesh sizes. Seeds are initialised by farthest-point sampling and refined over eight iterations; each vertex is assigned to the candidate seed minimising a combined distance $D = \sqrt{d_{\text{spatial}}^2 + (\beta\, d_{\text{normal}})^2}$, where d_{spatial} is the Euclidean distance normalised by the bounding-box diagonal, $d_{\text{normal}} = 1 - \mathbf{n}_v^\top \mathbf{n}_{\text{seed}}$ penalises normal disagreement, and $\beta = 0.3$ controls compactness. Superpixels are enforced to be spatially connected by reassigning disconnected fragments to neighbouring labels.

3.4 Graph Construction

Each superpixel becomes a node of an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, positioned at its centroid $\mathbf{c}_i \in R^3$ (mm) with associated patch normal $\mathbf{n}_i$. Every node carries a 5-dimensional feature vector $\mathbf{x}_i$ of rotation- and translation-invariant scalars: patch area, median mean curvature, median Gaussian curvature, normal dispersion ($1 - \|\bar{\mathbf{n}}\|$, capturing local roughness and branch take-offs), and local wall thickness. The last is measured landmark-free by casting a ray inward along $-\mathbf{n}_i$ and recording the distance to the opposite wall, replacing an earlier vessel-radius feature that had leaked the annulus–BCA axis. Curvature features are winsorised to the 1st–99th percentile to suppress mesh-defect spikes. Absolute coordinates and world-frame normals are deliberately excluded from $\mathbf{x}_i$: they reach the model only through the node positions used by the equivariant decoder. Edges connect superpixels that share a mesh boundary, each with a 3-dimensional invariant attribute $\mathbf{e}_{ij}$: centroid distance (mm), normal angle (degrees), and the number of shared boundary edges. Node and edge features are standardised using training-split statistics, which are stored with the model for inference.

3.5 Graph Attention Network with Equivariant Plane Heads

Node features are encoded by a three-layer GATv2 backbone [3], each layer applying four attention heads with averaged outputs, layer normalisation, and an ELU activation, and consuming the 3-dimensional edge attributes. This yields a per-node latent embedding $\mathbf{h}_i \in R^{64}$. To capture overall patient topology without introducing absolute-coordinate bias, a 128-dimensional global context vector $\mathbf{c} \in R^{128}$ is formed by concatenating global mean- and max-pooling representations of $\mathbf{h}$. Because the inputs are invariant descriptors, these embeddings are invariant to rigid transformations of the aorta.

Two independent rotation-equivariant heads decode the inlet and outlet planes. Within each head, a node-scoring MLP processes the concatenation $[\mathbf{h}_i \,\|\, \mathbf{c}]$ to produce scalar scores, which are normalised per graph by a softmax into attention

weights $w_i \in [0,1]$ with $\sum_i w_i = 1$. The plane origin is a soft-argmax convex combination of the superpixel centroids,

$$\hat{\mathbf{o}} = \sum_{i \in \mathcal{V}} w_i \, \mathbf{p}_i, \qquad \sum_{i \in \mathcal{V}} w_i = 1, \tag{1}$$

which localises the origin below superpixel resolution. The plane normal is derived from the attended surface normals through a weighted second-moment (covariance) matrix

$$\mathbf{M} = \sum_{i \in \mathcal{V}} w_i \, \mathbf{n}_i \mathbf{n}_i^{\top} + 10^{-6} \, \mathbf{I}, \tag{2}$$

where the small constant ensures numerical stability. The predicted normal $\hat{\mathbf{n}}$ is the unit eigenvector of $\mathbf{M}$ corresponding to its smallest eigenvalue: because radial wall normals vary least along the local vessel axis, this yields a stable normal even when attention concentrates on one side of the vessel. Both the origin (Eq. 1) and the normal (Eq. 2) are computed from invariant weights over geometric quantities, so rotating the input mesh transforms the predicted plane identically, i.e. the heads are rotation-equivariant.

3.6 Automated Clipping and Mesh Post-Processing

At inference the trained network predicts both planes in a single forward pass; the predicted origins are translated by the mesh centroid to restore absolute LPS coordinates. The surface mesh is clipped sequentially by the inlet and outlet planes, each normal oriented downstream toward the opposite plane. Floating connected components created by the planar cuts are removed by connectivity filtering anchored to the predicted annulus origin, and thin boundary artefacts are cleaned with a vertex-merge tolerance, retaining only the primary aortic segment. The result is a CFD-ready model with anatomically defined inlet and outlet boundaries, produced without manual landmark placement.

3.7 Implementation and Training Details

The model was implemented in PyTorch Geometric and trained on the QMUL Apocrita High Performance Computing (HPC) cluster using NVIDIA L40S GPUs [8]. The GATv2 backbone uses three layers with a hidden width of 64 and four attention heads per layer, producing a per-node embedding in R^{64} and a 128-dimensional global context vector. Optimisation used Adam with an initial learning rate of 1×10^{-3} and weight decay 1×10^{-4}, a cosine-annealing schedule decaying to 1×10^{-5} over 200 epochs, gradient-norm clipping at 1.0, and a batch size of 4 graphs. Node positions and target origins were zero-centred on each patient's mesh centroid, and node and edge features were standardised using statistics computed from the training graphs of each fold only, to prevent data leakage. Training ran for up to 300 epochs with an early-stopping patience of 40 epochs driven by the validation origin error. Owing to the limited cohort size, generalisation was assessed by leave-one-out cross-validation across the 15 cohort patients, together with a fixed held-out split reserving 5 unseen patients for independent testing.

Table 1. Plane differences and clipped-mesh similarity between the predicted and ground-truth results on the five held-out test patients. Origin: distance between plane centres; angle: tilt between cutting directions (sign-free); offset: displacement of the cut along the vessel; mean dist: average surface separation; HD95/HDmax: 95th-percentile and maximum surface separation; Dice: volume overlap.

	Inlet (annulus)			**Outlet (BCA)**			**Clipped-mesh similarity**			
Patient	origin (mm)	angle (deg)	offset (mm)	origin (mm)	angle (deg)	offset (mm)	mean dist (mm)	HD95 (mm)	HDmax (mm)	Dice
R_10	4.32	12.1	3.96	13.51	30.0	12.65	1.96	15.80	20.77	0.847
R_16	2.78	28.3	2.36	13.62	27.0	13.13	1.10	10.15	18.94	0.902
R_24	6.11	15.9	4.92	19.23	9.5	14.33	1.46	13.61	18.94	0.875
R_27	9.07	5.9	6.97	18.95	23.9	18.86	1.99	16.98	27.18	0.838
R_29	7.46	13.3	4.80	11.43	8.2	10.93	0.87	7.99	16.91	0.925
Mean	5.95	15.1	4.60	15.35	19.7	13.98	1.48	12.91	20.55	0.877
SD	2.22	7.4	1.49	3.15	9.1	2.67	0.45	3.38	3.53	0.033

4 Results

Evaluation Metrics. We evaluate the framework at two levels. First, *plane accuracy* measures how closely each predicted plane matches the expert-defined plane, using the origin distance (between plane centres), the angular error (the sign-free tilt between plane normals, in degrees), and the offset (the displacement of the cut along the ground-truth normal, in millimetres). Second, *mesh similarity* measures how closely the resulting CFD-ready geometry matches the expert-clipped mesh, using the symmetric mean surface distance, its 95th percentile and maximum (HD95, HDmax), and the Dice volume overlap.

Plane Accuracy. Table **??** reports the plane differences on the five held-out patients. The inlet (annulus) plane is located to 5.95 ± 2.22 mm, tilted by $15.1 \pm 7.4°$, with an offset of 4.60 ± 1.49 mm; the offset being smaller than the origin distance shows that part of the origin error is harmless in-plane sliding rather than a displaced cut. The outlet (BCA) plane is weaker and consistently so: 15.35 ± 3.15 mm origin, $19.7 \pm 9.1°$ angle, and 13.98 ± 2.67 mm offset. Here the offset is almost as large as the origin distance, so this is a genuine displacement of the BCA cut along the vessel; the small offset standard deviation (2.67 mm) indicates a systematic under- or over-reach at the branch rather than a single failure case. For a fully automatic method that places no landmarks, locating the annulus cut to within about 5 mm, comparable to the superpixel size and well within the aortic-root calibre, is anatomically plausible; the BCA is the clear limitation, as it is a small and variable structure supported by far fewer superpixels than the annulus.

CFD-Ready Mesh Similarity. Table **??** compares the clipped surface the planes produce against the expert-clipped mesh. The predicted surface lies on average 1.48 ± 0.45 mm from the reference, with a Dice overlap of 0.88 ± 0.03; the HD95 and HDmax are 12.9 ± 3.4 mm and 20.5 ± 3.5 mm. The low mean and

high Dice show that the clipped segment is correct almost everywhere: over the bulk of the vessel wall the two surfaces agree to one to two millimetres and the two solids overlap by nearly 90%, as seen in Figure 2. The much larger Hausdorff distances show that the disagreement is not spread out but concentrated in a small region, the outlet: the maximum separation of roughly 13–21 mm matches the outlet plane displacement of about 14 mm from Table **??**. Patient R_29 is the best case throughout (mean 0.87 mm, Dice 0.93), consistent with its small plane errors.

The two metric families are best read together. The mesh mean distance (1.5 mm) is much smaller than the plane origin errors, because most of the clipped surface is the vessel wall between the cuts, which both clips share; the plane error affects only the thin band near each cut and is diluted in a whole-surface average. Conversely, the mesh Hausdorff distance ($\sim$ 13 mm) recovers the outlet plane error almost exactly, because the worst-case measure is dominated by that same band. In short, the plane metrics say where and by how much the cut is wrong, while the mesh metrics say how much it matters for the delivered surface; the agreement between the outlet plane error and the mesh Hausdorff distance confirms that both measure the same underlying limitation from two directions.

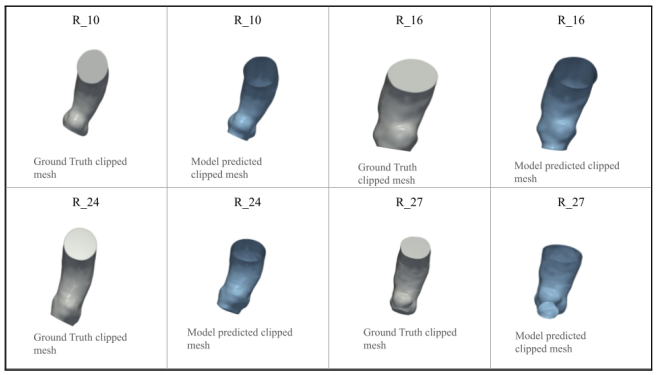


Fig. 2. Qualitative comparison across the five test patients, showing the expert-clipped (left) and predicted (right) meshes from the same viewpoint. The automatically clipped geometries closely follow the expert-defined surfaces, with the largest deviations confined to the clip boundaries.

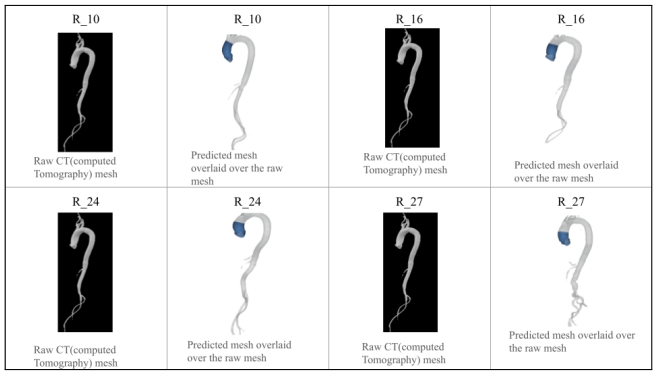


Fig. 3. Predicted clipping shown in anatomical context. For each sample, the full segmented aortic surface mesh derived from CT is shown alongside the same mesh with the automatically predicted clipped region overlaid in blue. The predicted region consistently isolates the aortic root and ascending aorta between the annulus and brachiocephalic artery, across variation in arch geometry and vessel tortuosity.

5 Conclusion

We presented a fully automated, landmark-free framework that predicts patient-specific clipping planes directly from segmented aortic surface meshes, removing the manual landmark placement that makes CFD preprocessing for TAVI planning slow and operator-dependent. The aorta is represented as a superpixel graph of rotation- and translation-invariant features, and a GATv2 network predicts the annulus and BCA planes through equivariant heads, so the prediction rotates with the anatomy and the same pipeline applies unchanged to a new, unannotated patient.

On the five held-out patients the produced geometries matched the expert-clipped meshes to a symmetric mean surface distance of 1.48 mm (Dice 0.88), with the inlet plane located to within 5.95 mm. The clearest limitation is the outlet: the BCA plane was displaced by about 14 mm on average, where the worst-case surface error concentrates, as the branch is small and supported by far fewer superpixels than the annulus. Future work will use the expanded Barts cohort to add training examples around the BCA, validate on a larger multi-centre dataset, and couple the generated domains to downstream haemodynamic simulation.

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References

1. Achanta, R., Shaji, A., Smith, K., Lucchi, A., Fua, P., Süsstrunk, S.: SLIC superpixels compared to state-of-the-art superpixel methods. IEEE Transactions on Pattern Analysis and Machine Intelligence **34**(11), 2274–2282 (2012)

2. Antiga, L., Piccinelli, M., Botti, L., Ene-Iordache, B., Remuzzi, A., Steinman, D.A.: An image-based modeling framework for patient-specific computational hemodynamics. Medical & Biological Engineering & Computing **46**(11), 1097–1112 (2008)
3. Brody, S., Alon, U., Yahav, E.: How attentive are graph attention networks? In: International Conference on Learning Representations (ICLR) (2022)
4. Du, P., An, D., Wang, C., Wang, J.X.: Ai-powered automated model construction for patient-specific cfd simulations of aortic flows. Science Advances **11**(35) (2025)
5. European Heart Journal Digital Health: Understanding the mechanisms of TAVI durability through computational modelling: A multidisciplinary review. European Heart Journal Digital Health (2026), doi:10.1093/ehjdh/ztag020
6. Fedorov, A., Beichel, R., Kalpathy-Cramer, J., Finet, J., Fillion-Robin, J.C., Pujol, S., Bauer, C., Jennings, D., Fennessy, F., Sonka, M., Buatti, J., Aylward, S., Miller, J.V., Pieper, S., Kikinis, R.: 3D Slicer as an image computing platform for the quantitative imaging network. Magnetic Resonance Imaging **30**(9), 1323–1341 (2012)
7. Hanocka, R., Hertz, A., Fish, N., Giryes, R., Fleishman, S., Cohen-Or, D.: MeshCNN: A network with an edge. ACM Transactions on Graphics **38**(4), 90 (2019)
8. King, T., Butcher, S., Zalewski, L.: Apocrita — high performance computing cluster for queen mary university of london (2017)
9. Martins, A.F.T., Astudillo, R.F.: From softmax to sparsemax: A sparse model of attention and multi-label classification. In: International Conference on Machine Learning (ICML). pp. 1614–1623 (2016)
10. Peters, B., Niculae, V., Martins, A.F.T.: Sparse sequence-to-sequence models. In: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 1504–1519. Association for Computational Linguistics (2019)
11. Qi, C.R., Su, H., Mo, K., Guibas, L.J.: PointNet: Deep learning on point sets for 3D classification and segmentation. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 652–660 (2017)
12. Queirós, S., Dubois, C., Morais, P., Adriaenssens, T., Fonseca, J.C., Vilaça, J.L., D'hooge, J.: Automatic 3d aortic annulus sizing by computed tomography in the planning of transcatheter aortic valve implantation. Journal of Cardiovascular Computed Tomography **11**(1), 25–32 (2017)
13. Rauseo, A., Chen, X., et al.: Computational modelling for patient-specific transcatheter aortic valve implantation: A systematic review. Frontiers in Digital Health **8** (2026)
14. Rauseo, A., Chen, X., et al.: Understanding the mechanisms of tavi durability through computational modelling: A multidisciplinary review. European Heart Journal Digital Health **7**(2), ztag020 (2026)
15. Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y.: Graph attention networks. In: International Conference on Learning Representations (ICLR) (2018)
16. Yao, L., Jiang, P., Xue, Z., Zhan, Y., Wu, D., Zhang, L., Wang, Q., Shi, F., Shen, D.: Graph convolutional network based point cloud for head and neck vessel labeling. In: Machine Learning in Medical Imaging (MLMI). Lecture Notes in Computer Science, vol. 12436, pp. 474–483. Springer (2020)